

The Effects of the 2018 Trade War on the Economy: A Review of the Literature

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Outline

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- III Employment
- IV Trade Reallocation
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I. Introduction

- ▶ 2018: Trump Admin imposes tariffs on wide range of products
- ▶ Several foreign governments respond with retaliatory tariffs
- ▶ 2019: 17 percent of U.S. imports covered by tariffs totaling over \$420 billion of U.S. imports in 2017
- ▶ 2021: Biden admin begins lowering or rescinding tariffs, but some remain, especially against China

I. Introduction

Customs duty revenue as a percent of goods imports

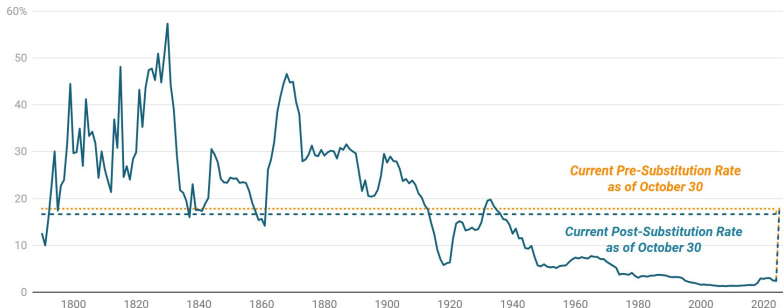


Chart: The Budget Lab • Source: Historical Statistics of the United States Ea424-434, Monthly Treasury Statement, Bureau of Economic Analysis, The Budget Lab analysis. • Created with [DatawrappR](#)

Figure: U.S. Average Effective Tariff Rate (Yale Budget Lab, 2025)

I. Introduction

- ▶ Commonly used regressions:

- ▶
$$\ln x_{ijt} = \eta_{ij} + \sum_{s=-T}^T \beta_s \left(1_{ijs} \times \ln \left(\frac{1 + \tau_{ijs}}{1 + \tau_{ij0}} \right) \right) + \delta_{jt} + \mu_{it} + u_{ijt}$$

- ▶ Common data sources:

- ▶ Harmonized Tariff Schedule, Bureau of Labor Statistics, Prices of products sold to consumers, U.S. Census Bureau, World Trade Organization, Compustat

- ▶ Interesting Measures

- ▶
$$\text{TECP}_{C,t} = \frac{\Delta_t \ln P_c(\Delta\tau)}{\left[\frac{M_{C,t-1}}{D_{C,t-1}} \right] \sum_k s_{C,t-1}^k \Delta\tau_{C,t}^k}$$

II. Welfare

- ▶ Foreign exporters did not lower their prices in response to tariffs, causing an almost complete pass-through to U.S. importers
- ▶ Welfare loss of about \$8 billion in 2018 → nearly \$25 billion by September 2019
 - ▶ 0.08% to 0.13% of real GDP
- ▶ Reduction in consumer surplus of nearly \$51 billion
- ▶ Producer prices increased (incomplete pass-through)

II. Welfare

Twelve-Month Proportional Change in Import Prices by Tariff Wave

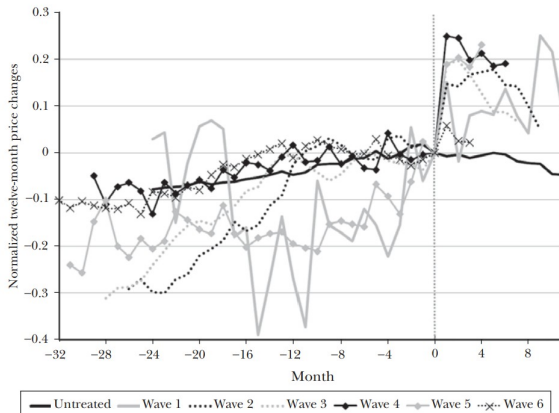


Figure: Change in Import Prices (Amiti et al., 2019)

II. Welfare

- ▶ Did U.S. exporters lower prices?
 - ▶ Amiti et al. (2019, 2020) and Fajgelbaum et al. (2020): No
 - ▶ Cavallo et al. (2021): Yes
- ▶ Passthrough to consumers
 - ▶ Rapid and nearly complete for some goods
 - ▶ Most goods saw incomplete (10% tariff resulted in a 0.35% increase)
 - ▶ Short term observation (front running and diversion)
- ▶ Some U.S. firms with tariff protection raised prices on their goods (washing machines and dryers)

II. Welfare

- ▶ Washer and dryer tariffs
 - ▶ Created 1,800 jobs
 - ▶ Total cost to consumers: \$1.55 billion annually
- ▶ Aggregate consumption fell by 0.28% to 0.39% (\$40 billion to \$54 billion)
 - ▶ Moving from least affected to most affected counties → 3.8% decrease

III. Employment

- ▶ Foreign retaliatory tariffs consistently found to decrease employment
- ▶ Import tariffs were found to have small or weak positive effects.
- ▶ Farm subsidies had a weak positive effect

III. Employment

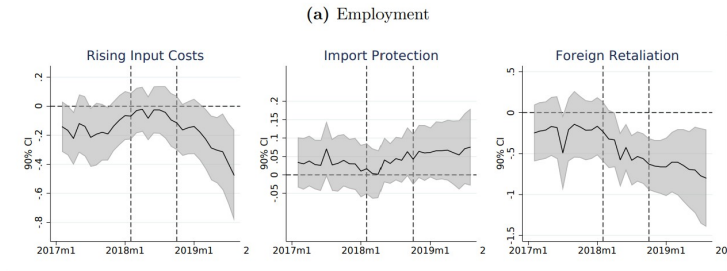


Figure: Cumulative Tariffs Effects (Detrended) Flaen and Pierce (2019)

III. Employment

- ▶ Total effect on employment?
 - ▶ Flaaen and Pierce (2019): Manufacturing Employment
 - ▶ Rising input costs and foreign retaliatory tariffs had a negative impact
 - ▶ Import protection had a positive effect, BUT outweighed by the other two measures
 - ▶ Autor et al. (2024): At best a wash, if not slightly negative after accounting for farm subsidies.
- ▶ Limitations
 - ▶ Short term effects & pre-trend violation
- ▶ Why?

IV. Trade Reallocation

- ▶ Autor et al. (2024): U.S. imports from China have relatively low trade cost elasticities; buyers were able to import goods from places without tariffs, not from domestic manufacturers.
- ▶ Cavallo et al. (2021) and Flaaen et al. (2020) show other cases where firms engaged in trade diversion. Was this consistent across the board?

IV. Trade Reallocation

- ▶ Trade Diversion
 - ▶ Imports from China of goods targeted by tariffs fell 26% while imports from the ROW increased by 4%
 - ▶ \$21 billion worth of decline in imports in first half of 2019 replaced by ROW.
- ▶ Front Running
 - ▶ Following announcement of trade investigation, imports from China rose by 16% and nearly 9% from ROW

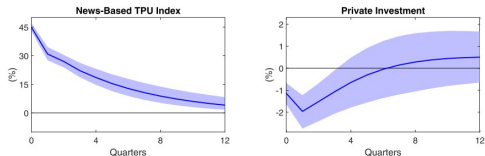
V. Investment

- ▶ For firms, reduction in capital stock $\rightarrow$ reduction in investment
- ▶ 2018: Firm level estimates of TPU suggest aggregate investment saw 1% decline
 - ▶ Vector auto-regression suggests up to 2% decline

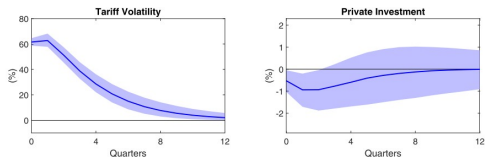
V. Investment

- ▶ Policy-induced stock-market declines → lower returns to capital → lowers investment
- ▶ Estimated 2% decline in investment growth rate by end of 2020
- ▶ U.S. and Chinese tariff announcements lowered U.S. aggregate equity prices in sample of firms by 6 percent, a \$1.7 trillion reduction in market value

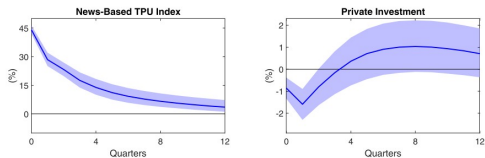
V. Investment



(a) BIVARIATE VAR WITH NEWS-BASED TPU



(b) BIVARIATE VAR WITH TARIFF VOLATILITY



(c) MULTIVARIATE VAR WITH NEWS-BASED TPU

Figure: Investment Effects of TPU and TVS Caldara et al. (2019)

VI. Imports & Exports

- ▶ Imports
 - ▶ First half of 2019
 - ▶ Decline of 26% of imports of targeted goods from China (relative to non targeted)
 - ▶ Decline of \$35 billion worth of imports
 - ▶ Decline in imports was not offset by trade diversion
 - ▶ \$14 billion worth lost (not replaced by ROW)
 - ▶ Less imports → less variety of goods available to consumers

VI. Imports & Exports

▶ Exports

- ▶ Waugh (2019): retaliatory tariffs cause a county's exports to fall
- ▶ Handley (2025): import tariffs caused exports to be 2.9 log points lower in 2019 relative to the pre-tariff period.
- ▶ Loss in exports equal to a tariff of 2 percent of the value of total exports.
- ▶ This ad valorem tariff reached 4% for the most exposed products

VI. Imports & Exports

- ▶ Benguria and Saffie (2025): Lost exports to China were rerouted elsewhere
- ▶ BUT exports affected by Chinese State Owned Enterprises were not
- ▶ At average SOE share, U.S. exports fell by 4 percent

VII. Conclusion

- ▶ Declines in
 - ▶ Welfare, imports, exports, consumption, investment
- ▶ Increases in
 - ▶ Front running, trade diversion, prices of some goods
- ▶ No change in
 - ▶ Employment (at best), most goods
- ▶ Limitations
 - ▶ Short term effects only (COVID makes long term effects extremely hard)
 - ▶ Small (but growing) literature

VII. Conclusion

- ▶ What about 2025?
 - ▶ Larger tariffs → disproportionately larger welfare losses
 - ▶ May see more consistent increases in prices
 - ▶ Larger decreases in investment due to increased volatility in actions made by U.S.
 - ▶ Opportunity to study long term effects